

# System Test Plan – Shapes Calculator

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## 1. Objective

To verify that the shape calculator behaves as intended and is conducive to a positive user experience. Any bugs with interactivity should be identified, including any issues with:

- Adding a shape
- Listing shapes
- Selecting a shape
- Performing calculations (perimeter & area)
- Exiting the application
- Navigating the console interface

## 2. Scope

Each piece of functionality will be tested for each type of shape, including instances with user input and default values for added shapes. Thus, six total shapes should be added and interacted with. The user experience with the program as a whole will also be observed.

## 3. Test Items

- Square 1 (*default, testing the constructor without parameters*)
- Square 2 (*user's choice of lengths, testing the constructor with parameters*)
- Rectangle 1 (*default, testing the constructor without parameters*)
- Rectangle 2 (*user's choice of lengths, testing the constructor with parameters*)
- Triangle 1 (*default, testing the constructor without parameters*)
- Triangle 2 (*user's choice of lengths, testing the constructor with parameters*)

## 4. Test Approach

The QA engineer has been instructed on the method and scope of this test. This individual will be given access to the application and must interact with it organically with the goal of covering all test items outlined above. Their behaviour and response to the application will be recorded, along with the technical outcomes of the program. Any issues with logic and bugs will be identified, along with feedback on the overall usability and user experience.

## 5. Test Cases

1. The application can open and be interacted with
2. For each shape, the default constructor can be used to add a shape with “default” values
3. For each shape, lengths can be supplied by the user to add a shape with non-default values
4. Shapes can be listed
5. Shapes can be selected
6. Perimeter can be calculated on a selected shape with default values
7. Perimeter can be calculated on a selected shape with non-default values
8. Area can be calculated on a selected shape with default values
9. Area can be calculated on a selected shape with non-default values
10. The application handles invalid input gracefully
11. The application can be closed gracefully
12. Menu navigation is possible

## 6. Expected Result

Each type of shape should be added to the calculator’s internal list, and perimeter and area calculations should be performed correctly on these shapes. The user should be able to list and view all added shapes and select one to perform calculations on. Perimeter and area calculations should be correctly evaluated for the selected shape. The program should feel easy to navigate with an intuitive menu. The program should not crash or quit unexpectedly.